Original Article

Acupoint Stimulation to Improve Analgesia Quality for Lumbar Spine Surgical Patients

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Abstract:

Lumbar spine surgery has a high incidence of postoperative pain, but this pain is treatable through many methods, including patient-controlled analgesia (PCA). Acupoint stimulation could be considered an adjunct to PCA, improving the effectiveness of analgesia for patients recovering from lumbar spine surgery. The current study aimed to examine the effect of acupoint stimulation with PCA on improving analgesia quality after lumbar spine surgery. A single-blinded, sham-controlled design was used for the experimental, not control, groups. Data collection for the control group was completed first, followed by data collection for the other 2 groups. Participants were randomly assigned to the acupoint stimulation (AS) (n = 45) or sham group (n = 45). All participants received structural PCA multimedia information before lumbar surgery. The AS group received auricular acupressure combined with transcutaneuos electric acupoint stimulation (TEAS) at the true acupoint; the sham group received acupoint stimulation in the same manner but at a sham acupoint and without embedding seeds; and the control group received no acupoint stimulations. The analgesia quality, analgesic consumption, and postoperative nausea and vomiting (PONV) were used as measure of effects for the interventions. Significant differences were found between the AS and control groups in pain intensity but not in the belief and satisfaction subscales of analgesia quality. Also found a significant difference among the 3 groups in analgesic consumption and the severity of PONV in the first 72 hours after surgery. The current study shows that the combination of auricular acupressure and TEAS reduced pain intensity, morphine consumption, and PONV severity. Acupoint stimulation could be considered a multimodal analgesia method and an adjunct to PCA for lumbar spine surgery patients.

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Pain control is a major problem for patients recovering from lumbar spine surgery (Yeh, Tsou, Lee, Chen, & Chung, 2010b). Although pain management in clinical practice has established policies and practices, approximately 50%–60% of lumbar spine surgical patients experience moderate pain at rest

(Gepstein, Arinzon, Folman, Shuval, & Shabat, 2007; Sommer et al., 2009). Postoperative pain contributes to bleeding (Guay, 2006), delayed recovery (Shiloh et al., 2003), emotional disturbance, and prolonged hospitalization (Kwekkeboom & Gretarsdottir, 2006). Suitable postoperative pain control positively influences wound healing and patients' postoperative satisfaction (Momeni, Crucitti, & De Kock, 2006). Therefore, reducing postoperative pain is essential.

Patient-controlled analgesia (PCA) is often used to administer opioids for postoperative pain management of lumbar spine surgical patients (Yeh, Chung, Chen, Tsou, & Chen, 2010a; Yeh et al., 2010b). According to a Cochrane systematic review of 55 randomized controlled trials, PCA provides better pain control and greater patient satisfaction than "as-needed" analgesia (Hudcova, McNicol, Quah, Lau, & Carr, 2006). However, patients undergoing spine surgery using PCA still experience moderate pain in the first 24 hours after surgery (Gepstein et al., 2007; Yeh et al., 2010a). Postoperative nausea and vomiting (PONV) is the most common complication of PCA (Wang, Shen, Liu, Xu, & Guo, 2009; Yeh et al., 2009). In addition, multiple factors inhibit optimal postoperative pain management, including inadequate preoperative education (Clarke et al., 1996), negative beliefs about pain (Tzeng, Chou, & Lin, 2006), and misconceptions about analgesics (Carr, 2007; Yeh & Chung, 2009; Yeh, Yang, Chen, & Tsou, 2007). Patient education can enhance pain awareness, reduce the barriers to effective pain management, and improve analgesia quality (Carr, 2007; Yeh et al., 2007). Providing PCA information to improve pain awareness and adding non-pharmacologic remedies are significant concerns in the management of postoperative pain, and may help reduce the adverse effects of opioid usage.

Auricular acupressure and transcutaneuos acupoint electric stimulation (TAES), which is a form of noninvasive acupoint stimulation, has shown positive effects on pain relief (Asher et al., 2010; Usichenko, Lehmann, & Ernst, 2008). According to the principle of Chinese medicine, the acupoints on the ear and meridian are closely tied to the internal organs (Yeh, Yang, Chen, & Tsou, 2004b). Acupoint stimulation is the application of stimuli to an acupoint to excite qi inside meridians (qi means a sensation that is often elicited to enhance the effect of acupuncture treatment; subjects experienced more aching, pulling, heavy, dull, electric, and throbbing sensations), stabilize the body, strengthen functions, and cure disease (Yeh, Chen, & Chen, 2011a). Clinical practitioners have introduced TAES, which intermittently stimulates acupoints by alternating low- and high-frequency electrical current into their practice. Using different frequencies, TAES releases different opioid peptides that produce analgesic effects in the nervous system. Low-frequency stimulation, such as 2 Hz (Hertz, the unit of sound frequency; one Hz is equal to one cycle per second.), induces analgesia by releasing β-endorphins, enkephalin, and orphanin, whereas high-frequency stimulation, such as 80–100 Hz, releases dynorphin (Lin & Chen, 2008; White, Cummings, & Fishie, 2008). Previous research has shown that TAES produces a significant decrease in PCA morphine consumption after lumbar spine surgery (Yeh et al., 2010a).

PCA pain treatment is associated with morphine-related side effects, and previous research has suggested how to minimize such effects (Yeh, Chung, Chen, & Chen, 2011b). Lumbar spine surgery is a complex operation, and commonly taking 200 to 270 minutes (min) (Yeh et al., 2010a; Yeh et al., 2010b). Lumbar spine surgery patients with PCA feel the worst postoperative pain at a moderate intensity during the first 72 hours after surgery. However, auricular acupressure alone does not relieve pain efficiently (Yeh et al., 2010b). In contrast, TAES has analgesic effects during the first 24 hours when used alone (Yeh et al., 2010a) or when combined with acupuncture (White et al., 2008). The effect of acupressure alone was insufficient to relieve postoperative pain. In addition, the adjuvant effects of the combination of auricular acupressure and TAES for lumbar spine surgical patients with PCA require further research-based support.

AIM

The current study aimed to examine the effects of acupoint stimulation on postoperative analgesia quality, analgesic consumption, and PONV severity during the first 72 hours after lumbar spinal surgery. There were significant differences in postoperative analgesia quality, analgesic consumption, and PONV severity were hypothesized among three groups.

METHODOLOGY

Design

A single-blinded, sham-controlled design was used for the experimental, not control, groups. To avoid contamination of intervention, participants were enrolled in tandem, whereby data collection for the control group was completed first, followed by data collection for the other two groups. Computer-generated numbers were used, and information on the intervention was allocated sequence in opaque, sealed envelopes to ensure random assignment of participants to the AS or sham group. Each participant received the information through PCA multimedia. The sham acupoint stimulation was evaluated as a placebo effect. Individual participants and medical staff did not know which group participants were assigned to. The acupoint stimulation (AS) group received acupoint stimulation at true acupoints. The sham group received acupoint stimulation in the same manner, but at sham acupoints with low-frequency current or at true acupoints without pressure. The control group received no acupoint stimulation.

Participants

All participants were diagnosed with spinal stenosis, spondylolithesis, and a herniated intervertebral disc, according to the pathological examination, and underwent lumbar spine surgery. Lumbar surgery patients were recruited from the orthopedic ward of a 2,909-bed medical center in Taiwan. Inclusion criteria were an age of 18 years and older, general anesthesia, American Society of Anesthesiologists (ASA) Physical Status Classification System ≦ III (a patient with severe systemic disease), consent to PCA, and directly returning to the ward from the anesthesia recovery room. Exclusion criteria were the use of antiemetics before surgery, usage of a pacemaker, history of arrhythmia or epilepsy, opiate dependence, clinically significant cardiovascular diseases, and abnormally shaped earlobes or cutaneous lesions at the application sites. Based on the primary outcome of morphine consumption with a medium-effect size ($f = 0.25$) at 5% level of significance with 80% power using 3-times repeated measures (Cohen, 1988), a minimum of 108 participants were required for GPower 3.1.3. The population was oversampled by 25% to account for an expected follow-up loss of approximately 25%; 135 participants would be necessary. Figure 1 shows a flowchart of the research design and participants of this study.

Ethical Considerations

The study hospital's human research ethics committee approved the study, and obtained written informed consent from all participants after explaining the study. No safety issues were anticipated for this study. Participants were also made aware that all data remained confidential at all times, and that they were free to withdraw at any time during the study without affecting their care.

Interventions

The PCA device, with 100 milligram (mg) morphine and 2 milliliter (mL) droperidol per 100 mL, was connected to the patient's intravenous line within 15 minutes of admittance to the anesthesia recovery room and continuously administered for 72 hours. The loading dose was 2.0–3.2 mg, the basal infusion was 0.40–0.75 mg h^{-1}, and the bolus dose was 0.7–1.0 mg, with a 6-minute lockout interval between each PCA dose and 4 hours maximal dosage of 8–10 mg. To control the influence of inappropriate PCA use, all participants received instructions on how to use a PCA device by PCA multimedia and printed booklets the evening before the surgery. The major content of the PCA multimedia videodisc and printed booklet included 4 main sections: Introduction to PCA, Condition, Nursing Guide, and Frequently Asked Questions (FAQs) about clinical conditions or situations related to pain (Yeh et al., 2004b). Each section included word descriptions, pictures, figures, films, and sound. The length of the entire program was approximately 20 minutes of continuous play. Previous research has shown that this PCA multimedia program and printed booklet achieves significant improvement in pain awareness and analgesia quality (Yeh et al., 2007). The PCA multimedia and printed booklets were given to each participant followed by a one-on-one discussion to ensure that the participants could use the PCA device correctly.

Auricular acupressure was applied to 5 common auricular acupoints: shenmen (TF4), lumbosacral vertebrae (AH10), kidney (CW8), subcortex (AT5), and stomach (CO4). Shenmen raises endorphin levels in the body, thus reducing pain (Yeh, Chen, & Lin, 2004a), lumbosacral vertebrae is the most commonly used point for clients with low back pain because it reduces lumbosacral pain (Suen, Wong, Chung, & Yip, 2007), kidney strengthens the lumbar (Suen et al., 2007), and subcortex and stomach normalize qi energy flow and harmonize the stomach (Chou & Chou, 2008). For the AS group, a seed-embedding method with Wangbuliuxingzi seeds (Beijing, China) was used to prolong stimulation at the auricular acupoint. Pressure was applied by thumb and index finger for 3 minutes per acupoint for a total session duration of 15 minutes (5 acupoints x 3 min). Ten treatments were provided over the course of the study, including 1 hours and 3 hours after returning to the ward and 4 times at Day 1 and Day 2 after surgery. The de qi sensation appears to be an established intervention efficacy indicator during auricular acupressure and TAES (Han, 2004). In addition, TAES was applied to the Weizhong (BL40) and Yanglingquan (GB34) acupoints (Lin, 2010). A transcutaneous electrical muscle/nerve stimulator (GP8016M, Sanidad Tens/Ems, FDA510K: K021430) with 2 pairs of disposable electrode pads (3 x 3 cm) was established with an alternating low- and high-frequency current with a pulse duration of 0.25 milliseconds for 20 minutes at 1 hours and 3 hours after returning to the ward. The intensity ranged from

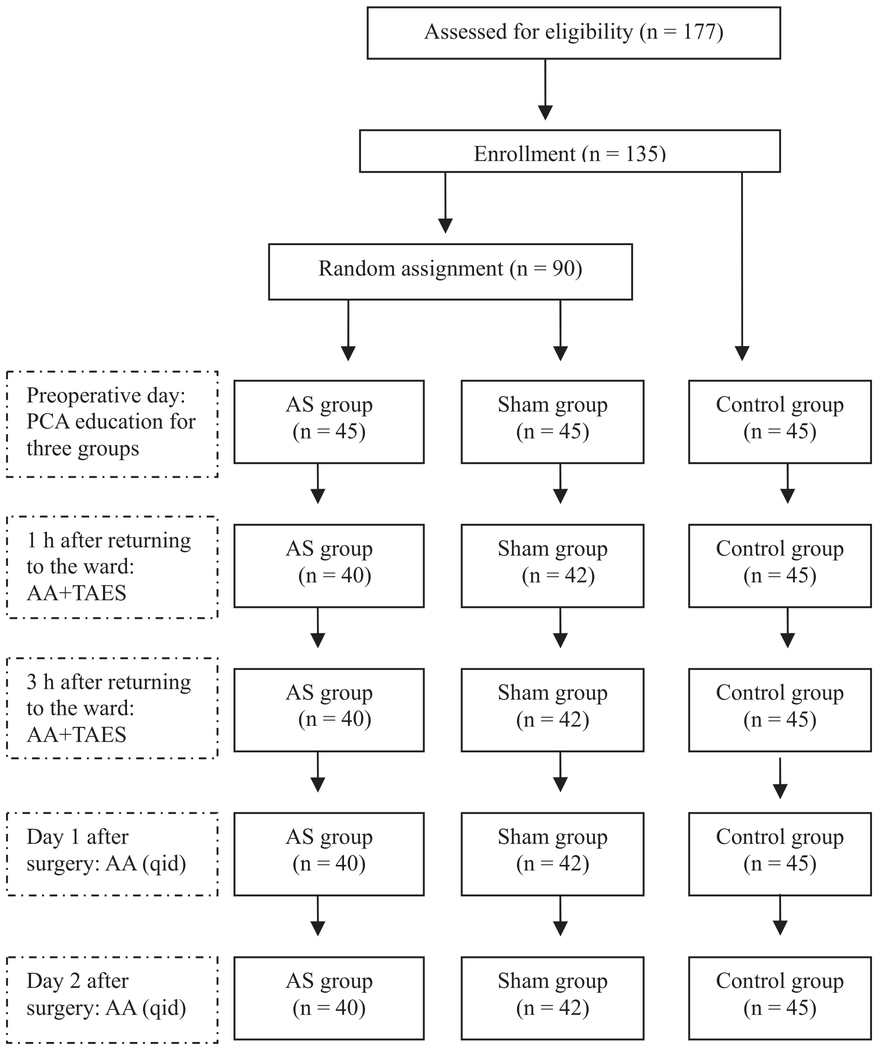

Figure 1. ■ The flow chart of research design, participants' recruitment, intervention, and assessment. AS = acupoint stimulation; AA = auricular acupressure; TAES = transcutaneuos acupoint electric stimulation.

4 to 7 milliamps (mA), depending on the participant's tolerance. The sham group received an intervention in the same manner, but acupressure was applied at the true auricular acupoint without embedding seeds, and TAES was applied at a sham acupoint (2 cm away from the true acupoint and not on any meridians) with a frequency current of less than 1 mA. The control group received no acupoint stimulation. Interventions were provided by a research nurse who was a qualified member of the research team, and verified by two traditional Chinese medicine physicians.

Data Collection

Demographic data and intraoperation parameters included ASA class, anesthesia time, operating time, anesthesia time, intraoperative blood loss, and recovery room time. The American Pain Society Patient Outcome Questionnaire (APSPOQ) was used to evaluate the analgesic quality regarding the brief pain inventory (BPI), satisfaction, and belief subscales (Yeh et al., 2007). These BPI items address pain intensity, and ask patients to rate their current pain, worst pain, and average pain over the past 24 hours. The other 5 items ask patients to rate the level at which their pain interferes with certain functions such as activity, mood, deep breathing and coughing, sleep, and relations with others (Yeh et al., 2007). Pain intensity was assessed using the visual analogue scale (VAS) on a scale ranging from 0 (no pain) to 10 (very severe pain). The pain interference with functionality was scored from 0 to 10. The satisfaction subscale consists of 3 items, using a 6-point scale that ranges from very dissatisfied to very satisfied. The belief subscale

TABLE 1.
Homogeneity of the Demographics and Intraoperative Parameters Among Groups

Variables	Acupoint Stimulation (n = 40)	Sham (n = 42)	Control (n = 45)	*F* or χ^2 (*p*)
Age (years, Mean ± *SD*)	60.25 ± 13.42	59.79 ± 12.61	54.24 ± 14.63	*F* = 2.62 (0.08)
Gender (n, %)				
Female	30 (75.0)	26 (61.9)	22 (51.1)	χ^2 = 5.14 (0.08)
Male	10 (25.0)	16 (38.1)	22 (48.9)	
Body mass index (kg/m^2, Mean ± *SD*)	25.70 ± 4.36	26.70 ± 3.92	26.23 ± 4.97	*F* = 0.51 (0.60)
Anesthesia history (n, %)				
0	11 (27.5)	19 (45.2)	21 (46.7)	*F* = 4.46 (0.34)
1	20 (50.0)	17 (40.5)	15 (33.3)	
≧2	9 (22.5)	6 (14.3)	9 (20.0)	
Preoperative pain (Mean ± *SD*)	4.23 ± 2.43	4.16 ± 1.80	4.11 ± 1.81	*F* = 0.04 (0.96)
American Society of Anesthesiologists (n, %)				χ^2 = 3.08 (0.56)
I	6 (15.0)	5 (11.9)	11 (24.4)	
II	28 (70.0)	31 (73.8)	30 (66.7)	
III	6 (15.0)	6 (14.3)	4 (8.9)	
Anesthesia time (min, Mean ± *SD*)	260.75 ± 60.06	275.60 ± 81.92	259.93 ± 81.12	*F* = 0.54 (0.58)
Operating time (min, Mean ± *SD*)	212.90 ± 65.17	231.71 ± 74.39	215.00 ± 80.06	*F* = 0.82 (0.44)
Intraoperative blood loss (mL, Mean ± *SD*)	477.00 ± 309.19	598.68 ± 328.69	469.33 ± 300.00	*F* = 2.14 (0.12)
Recovery room time (min, Mean ± *SD*)	169.15 ± 40.31	160.60 ± 33.45	161.76 ± 40.93	*F* = *0.59* (0.56)

consists of 7 items that address patients' perceptions of barriers to or beliefs about pain management (cannot control pain, addiction, good patients avoid talking about pain, side effects, complaining about pain distracts physicians from treating underlying illness, tolerance, and progression of disease). Participants rated belief items using a 6-point scale ranging from 0 (do not agree at all) to 5 (agree strongly). The content validity index of the pain intensity, pain interference, satisfaction, and belief scales were .84, .78, .55, and .76 respectively (Dihle, Helseth, & Christophersen, 2008), and the corresponding Cronbach α values were .84, .74, .44, and .72. The analgesic consumption included PCA with morphine, oral analgesic medication, and analgesics for the first 72 hours after surgery. Equianalgesic doses based on a conversion chart for opioid analgesics were calculated (Donnelly, Davis, Walsh, Naughton, & World Health Organization, 2002). The Chinese version of the Index of Nausea, Vomiting and Retching (INVR) was used to evaluate PONV (Fu & Rhodes, 2002). This evaluation, which consists of self-evaluation and observation, covers times, severity, duration, amount of vomiting, and degree of discomfort. Participants rated each item in the INVR inventory from 0–4 with the total possible score of 32 (whatever the maximum score could be if all items rate as 4). Higher scores indicate more severe symptoms. The Cronbach alpha values of internal consistency reliability was .94–.95 (Fu & Rhodes, 2002).

Data Analysis

Data were analyzed by IBM Statistical Package for the Social Sciences (SPSS) 20.0 for Windows. Demographic data and intraoperative parameters were used as well as descriptive statistics and the chi-square test to verify homogeneity among the groups. Both the postoperative pain intensity and pain interference scores at repeated measures violated the assumption of normal distribution. Therefore, the difference in effects of outcomes among the groups was analyzed with generalized estimating equations (GEE).

RESULTS

One-hundred-seventy-seven participants met the study eligibility requirement; 135 signed the informed consent and were accepted into the study. Of this sample, 127 completed the study. Eight withdrew (7 refusing AS, and 1 having a fever), for an attrition rate of 6%. The age of participants ranged from 19–82 years (mean = 57.97 ± 13.78), and 62.2% (n = 79) were women. Table 1 presents a summary of the demographic and intraoperative parameters. There were no differences among the 3 groups in terms of demographic and intraoperative parameters ($p > .05$).

Figure 2 shows the pain intensity VAS score of BPI over time for the AS, sham, and control groups. The pain intensity scores at the first 24 hours after surgery were significantly different among the groups ($F = 4.81$; $p = .01$) with a significant result of the Scheffe

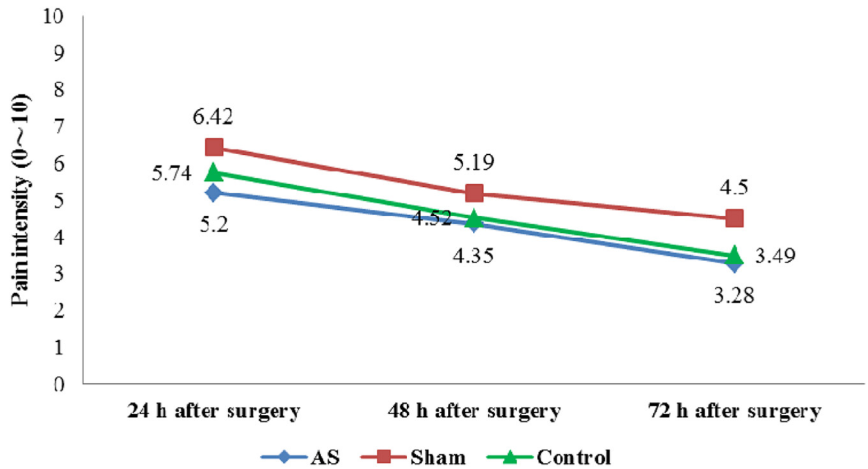

Figure 2. ■ Trends in pain intensity over time.

post hoc test between the AS and sham groups ($p = .01$). Figure 3 shows that pain interferes with function in a trend over time for 3 groups. The pain interference scores at the first 24 hours after surgery were significantly different among the groups ($F = 4.95$; $p = .009$) with a significant result of the Scheffe post hoc test between the sham and control groups ($p = .01$).

Table 2 presents a summary of the GEE results of pain intensity, pain interference, and PONV, including estimates and standard errors for a model with a correlation matrix and model-based estimates of variance. After adjusting for age and gender, this model showed significant between-group differences over time in pain intensity for the AS group compared to control group ($p = .02$) but not between the sham and control groups ($p = .09$). After surgery, the time effect was significant during 24–48 hours ($p < .001$) and 48–72 hours ($p < .001$) compared to 0–24 hours. Trend differences (interactions between time and group) showed no significant differences ($p > .05$) in pain intensity at 24–48 hours versus 0–24 hours and 48–72 hours versus 0–24 hours for the AS and sham groups. After adjusting for age and gender, the model showed a significant between-group difference over time in pain interference with functionality for the sham group compared to the control group ($p = .005$) but not between the AS and control groups ($p = .96$). After surgery, the time effect was significant during 24-48 hours ($p < .001$) and 48-72 hours ($p < .001$) compared to 0–24 hours. Trend differences (interactions between time and group) showed no significant differences in pain interferences with functionality at 24–48 hours versus 0–24 hours and 48–72 hours versus 0–24 hours for the AS and sham groups.

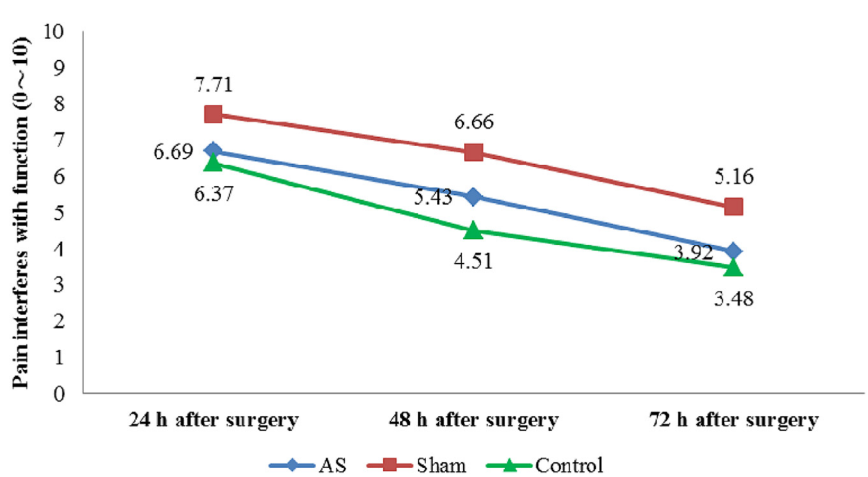

Figure 3. ■ Trends in pain; interferences with pain over time.

TABLE 2.
Results of Generalized Estimating Equations in Pain Intensity, Pain Interference, and PONV

Parameters	Pain intensity[a]		Pain interferences[a]		PONV[b]	
	Estimate	*p*	*Estimate*	*p*	*Estimate*	*p*
Intercept	5.79	< .001	5.81	< .001	1.31	.04
AS group*	−0.76	.02	0.02	.96	2.13	.02
Sham group*	0.56	.09	1.17	.005	2.83	.002
24∼48 h[c]	−1.22	< .001	−1.86	< .001	−0.73	.41
48∼72 h[c]	−2.26	< .001	−2.89	< .001	−0.91	.31
AS group×24∼48 h	0.37	.44	0.63	.29	−2.49	.10
AS group×48∼72 h	0.33	.48	0.16	.79	−2.11	.06
Sham group×24∼48 h	−0.004	.99	0.81	.17	−1.11	.39
Sham group×48∼72 h	−0.34	.46	0.34	.56	−2.14	.10

PONV = postoperative nausea and vomiting; AS = acupoint stimulation;
[a]Adjusted age and gender, [b]Adjusted antiemetics (yes/no), *Referenced group: the control group, [c]Referenced group: 24 hours after surgery, ×: interaction.

At 72 hours after surgery for the AS, sham, and control groups, the belief subscale scores were 2.11 ± 0.80, 2.11 ± 0.83, and 2.16 ± 0.87 respectively with no significant difference among the three groups ($F = 0.05$; $p = .96$). In addition, the satisfaction subscale scores were 5.46 ± 0.64, 5.39 ± 0.63, and 5.16 ± 0.73 respectively with no difference among the three groups ($F = 2.67$; $p = .08$). In addition, in the first 72 hours after surgery for the AS, sham, and control groups, the overall equianalgesic morphine consumption values were 69.16 ± 25.62 mg, 84.36 ± 28.93 mg, and 94.16 ± 29.89 mg respectively with a significant difference among the three groups ($F = 8.34$; $p = .001$) and in the Scheffe post hoc test between the AS and control groups ($p = .001$). The numbers of participants who used the antiemetics were respectively 3, 4, and 4.

The mean score of PONV from INVR data at 24 hours, 48 hours, and 72 hours after surgery was respectively 3.70, 0.38, and 0.68 for the AS group, 4.48, 2.64, and 1.43 for the sham group, and 1.62, 0.89, and 0.71 for the control group. Table 2 presents a summary of the GEEs result of PONV, including estimates and standard errors for a model with an independence correlation matrix and model-based estimates of variance. After adjusting for antiemetics (yes/no), the model showed significant between-group differences over time in PONV for the AS ($p = .02$) and sham groups ($p = .002$) respectively compared to the control group. After surgery, the time effect was not significant during 24–48 hours ($p = .10$) and 48-72 hours ($p = .06$) compared to 0-24 hours. Trend differences (interactions between time and group) showed no significant differences ($p > .05$) in PONV at 24–48 hours versus 0–24 hours, and 48–72 hours versus 0–24 hours for both the AS and control groups.

DISCUSSION

The current study presents the adjuvant effects of auricular acupressure combined with TEAS on reducing postoperative pain intensity, analgesic consumption, and PONV severity for lumbar spine surgical patients. The mean values of pain intensity show that the acupoint stimulation achieved efficient pain control during the first 72 hours, but this decreased over time and was attenuated by Day 3 after surgery. This is the first study to compare auricular acupressure combined with TEAS with non-acupoint stimulation used as an adjuvant in reducing postoperative pain after lumbar spine surgery. The acupoint stimulation in this study showed a true, not placebo, effect in lowering pain. A similar finding was presented by Yeh, et al. (2009) who showed that TEAS is beneficial in reducing pain intensity, average pain, and worst pain experienced in the first postoperative 24 hours. Chao et al. (2007) showed the positive effects of electrical stimulation in reducing pain in the first labor stage. Some studies have also supported electro-acupuncture on the effects of pain reduction for abdominal surgery (El-Rakshy, Clark, Thompson, & Thant, 2009), cardiac surgery (Coura, Manoel, Poffo, Bedin, & Westphal, 2011), and inguinal hernia repair (Dias et al., 2010) but not nasal septoplasty (Sahmeddini, Farbood, & Ghafaripuor, 2010). These contradictory results may have resulted from insufficient stimulation. A 20-30 minutes electrical stimulation period for pain reduction is suggested to improve intervention effects (Chao et al., 2007; Coura et al., 2011; Dias et al., 2010).

In the current study, the level of pain interfering with functionality was similar at any time after surgery for the participants in the AS and control groups but not for the sham group. Figures 2 and 3 showed that the participants receiving the sham acupoint stimulation had the most pain intensity and pain interference in the first 72 hours after surgery. When pain is not well controlled, it is a major impediment to postoperative functional recovery. Pain is also a persistent problem in lumbar spine surgery, such as for walking and changing positions (Yeh et al., 2010a). Most participants had positive beliefs about pain control, and this might have been caused by receiving PCA information before surgery to enhance correct awareness of pain control. This is consistent with the findings by Yeh et al. (2007) who showed that PCA multimedia intervention is effective in enhancing learning and pain awareness. Therefore, the analgesia quality was improved. Most participants were also satisfied with pain management. In a review study by Gordon et al. (2002), all 20 studies reported that although patients were satisfied with pain management, they also had high levels of pain. Yeh et al. (2010a) also reported a similar finding in lumbar spine surgical patients. The current study may provide evidence of the effectiveness of PCA information in increasing satisfaction with postoperative pain management. It is important for healthcare providers to consider patient satisfaction when attempting to improve adherence to pain management strategies in clinical practice. In conclusion, postoperative pain was not relieved optimally in patients with PCA for the sham and control participants in the first 72 hours after lumbar spine surgery, indicating beneficial aspects of acupoint stimulation for participants in the AS group.

In addition to pain reduction, acupoint stimulation also reduced analgesic consumption in the first 72 hours after surgery. This is consistent with the findings of other studies (Chen et al., 1998; Wang et al., 1997; Yeh et al., 2010a). Chen et al. (1998) reported a 39% reduction of morphine consumption by TEAS in lower abdominal surgical patients, whereas the pain was similar to sham-TEAS, non-acupoint-TEAS, and dermatomal-TEAS treatment groups. Wang et al. (1997) found that high TEAS caused a 30%–35% reduction in the PCA opioid requirement, and 90%–95% of patients reported adequate postoperative pain treatment under the same pain level as PCA only, PCA plus sham-TEAS, and low TEAS treatment groups. In addition, auricular acupuncture significantly reduced analgesic consumption for total hip arthroplasty (Usichenko et al., 2006) and ambulatory knee surgery (Usichenko et al., 2007) at a postoperative pain level similar to the non-acupuncture treatment group. Such studies used the same strategy of preoperative education about PCA, and acupressure/acupoint stimulation may be essential in decreasing postoperative pain, nausea, vomiting, and analgesic consumption.

This study shows that acupoint stimulation reduced PONV severity during the first 72 hours after surgery. This result is not in agreement with the findings by Yeh et al. (2010a) who showed that 30 minutes of acupoint electrical stimulation repeated twice does not reduce PONV severity in the first postoperative 24 hours. This might be because Yeh et al. only followed up the first 24 hours after surgery and measured it at one time point. The current study further analyzed PONV scores in the same time period of the first 24 hours and resulted in no difference among groups ($F = 2.47$, $p = .09$). Lower PONV scores on the first operative day may be associated with a lower use of PCA if the morphine dosage is the greatest predictor of PONV (Apfel, Läää, Koivuranta, Greim, & Roewer, 1999). PCA administration is usually by intravenous injection within 72 hours after surgery, and the most common morphine-related side effect was vomiting/nausea (Chen et al., 1998; Wang et al., 1997). Data should therefore be collected at least in the first 72 hours after surgery and repeated, which could provide trends of opioid doses and time effects of acupoint stimulation over time. Previous studies have shown that transcutaneous acupoint stimulation can reduce the incidence of vomiting/nausea but have not investigated the severity of vomiting/nausea. In the current study, auricular acupressure was applied to the subcortex and stomach acupoint to normalize qi energy flow and harmonize the stomach. This may be akin to adding antiemetics to PCA, but further research is warranted. For the AS, sham, and control groups, PCA morphine consumption was respectively 50.30 mg, 60.22 mg, and 70.31 mg, and the total amount of droperidol consumption was 2.52 mg, 3.01 mg and 3.52 mg. Compared to the AS group, the control group used much more antiemetic therapy. Thus, a decrease in PONV severity is a true effect of AS.

Limitations

This study has several limitations. First, the authors used a randomized controlled trial in addition to a non-randomized cohort; therefore, a selection bias may exist. Second, the sample from one hospital may not be generalizable. Future investigations should involve multiple sites. Exploring the relevant effects of AS without a PCA population of lumbar spine surgery is also suggested.

CONCLUSION

The current study demonstrated the effects of acupoint stimulation for lumbar surgical patients in reducing pain intensity, postoperative analgesic requirements, and severity of PONV. Acupoint stimulation is free from the complications of needle acupuncture when utilized as an adjunct to PCA after lumbar spine surgery. Additionally, the combination of auricular acupressure and TAES is a feasible, practical, and safe method for analgesia in lumbar spine surgical patients. Therefore, the current study suggests the combination of auricular acupressure and TAES could be used as an adjunct to PCA on improving analgesia quality for lumbar spine surgical patients.

Implication for Nursing

Postoperative pain is still a major concern of many patients. Despite the limitations, the findings of the current study add to the evidence on the benefits of using auricular acupressure combined with TEAS for the reduction of postoperative pain intensity, opioid consumption, and PONV severity. Such effects on integrating traditional Chinese medicine into western medicine for managing postoperative pain and opioid-related side effects can provide valuable references for clinical practices. Non-invasive acupoint stimulation, TEAS and acupressure, is easy and safe to be applied to nursing care for improving postoperative analgesia quality. This specific recommendation for surgical nurses is not only useful but it is also appreciated by patients.

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